



**Creo® Parametric
VB API Release Notes**
12.4.0.0

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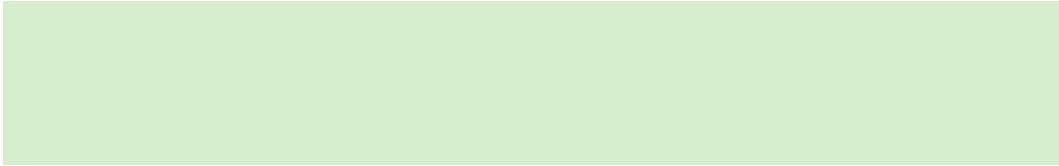
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About This Guide

This document provides release-specific information for VB API for Creo Parametric 3.0. It describes new methods and properties in VB API and also includes critical functions that require investigation of existing applications and adjustment of existing code.

Documentation for VB API

VB API documentation comes in the following forms:

- API Wizard in HTML format
- *VB API User's Guide* in PDF format
- **Reference Documents** page at <http://www.ptc.com/appserver/cs/doc/refdoc.jsp>

(Choose the product or document type)

The API Wizard and the *VB API User's Guide* are on the Creo Parametric DVD-ROM. You can view the API Wizard with your Internet browser. To view and print the PDF book, you must have Acrobat Reader installed. PTC welcomes your suggestions and comments on its documentation. Send comments electronically to:

MCAD-documentation@ptc.com

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Summary of Technical Changes

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 and VB API. It also lists the new and superseded functions for this release.

Critical Technical Changes

This section describes the changes in Creo Parametric 4.0 and VB API that might require alteration of existing VB API applications.

Accessing Member Information in a Pattern of Pattern

From Creo Parametric 2.0 M170, the method `IpfcFeaturePattern.ListMembers()` returns all the pattern header features created at the first level for a pattern of pattern. It does not return the pattern members under each pattern header.

Non-Applet Based Version of the APIWizard

To open the non-applet based version of the VB API APIWizard, point your browser to:

```
<creo_vbapi_loadpoint>\vbapidoc\index.html
```

A page containing links to the VB APIWizard and User's Guide will open in the web browser.

The non-applet based version of the VB APIWizard has enhanced search capabilities. The new search options enable you to search for only enumeration and exceptions, in addition to searching for all classes and functions.

Applet Based Version of the APIWizard

From Creo 4.0 F000, the Applet Based APIWizard is no longer supported. Use the non-applet based APIWizard instead.

Support for Visual Studio

From Creo Parametric 4.0 F000, VB API will support both Visual Studio 2012 and Visual Studio 2015. In a future release, Creo Parametric will fully support Visual Studio 2015. When Creo Parametric fully supports Visual Studio 2015, VB API will be available only with Visual Studio 2015.

Change in Behavior of IpfcServer. IsObjectCheckedOut()

From Creo Parametric 2.0 M040 onward, the behavior of the method `IpfcServer.IsObjectCheckedOut()` has been fixed. The method now returns the value `false` to indicate one of the following statuses for the specified object:

- the object was only saved, but never uploaded
- the object was only uploaded, but never checked in

Change in Behavior of IpfcTable. CheckIfIsFromFormat()

The method `IpfcTable.CheckIfIsFromFormat()` did not correctly check if the drawing table was created using the drawing format. The method would return `false`, if the first segment of the table was not on the current sheet in the Creo Parametric user interface, even though the table was created using the drawing format. This behavior has been fixed in Creo Parametric 2.0 M120. The method now checks and returns the correct boolean value.

As in the previous releases, the method `IpfcTable.CheckIfIsFromFormat()` ignores the value provided in the input argument `SheetNumber`.

Change in Directory Structure for Creo Installation

From Creo 3.0 onward, the directory structure for Creo installation has changed from:

- `<creo_loadpoint>\Parametric` to `<creo_loadpoint>\<version>\Parametric`.
- `<creo_loadpoint>\<version>\Common Files` to `<creo_loadpoint>\<version>\Common Files`

Change in Integer Values for Enumerated Data Type EpfcComponentType

The integer values for enumerated data type `EpfcComponentType` were shifted by 1 and did not reflect the actual values. From Creo Parametric 2.0 2.0, M120 onward, this behavior has been fixed. If your application code uses these values, you must rebuild your application.

Digital Rights Management Retired

From Creo Parametric 3.0 onward, Digital Rights Management (DRM) is no longer supported. VB API applications that check the DRM permissions will have to be updated.

Disable Notification Messages in Trail Files

When notifications are set in VB API applications, every time an event is triggered, notification messages are added to the trail files.

From Creo Parametric 2.0 M210 onward, a new environment variable `PROTK_LOG_DISABLE` allows you to disable this behavior. When set to `true`, the notifications messages are not added to the trail files.

Documentation Updated for `IpfcServer.IsObjectModified()`

The documentation for the method `IpfcServer.IsObjectModified()` has been updated in Creo 3.0 M010. This method returns `true` if the object was modified locally.

Layout Model Type

From Creo 3.0 M010 onward, the enumerated type `IpfcModelType` contains an additional value `EpfcMDL_CE_SOLID` that represents models of type Layout. VB API methods will only be able to read models of type Layout, but will not be able to pass Layout models as input to other methods. To work with Layout models, you must rebuild your existing VB API applications.

No Support for Boundary Box Type of Simplified Representation

From Creo Parametric 3.0 onward, the boundary box type of representation specified by the following values are no longer supported:

- `EpfcSIMPREP_BOUNDBOX_REP` in the enumerated data type `EpfcSimpRepType`
- `EpfcSIMPREP_BOUNDBOX` in the enumerated data type `EpfcSimpRepActionType`

Retrieving Solids in a Creo Parametric Session Linked to Windchill

You must retrieve solid models in a Creo Parametric session, which is linked to Windchill, only after you create a new workspace. If you retrieve the models before creating the workspace, the models would be erased from the Creo Parametric session.

Support for Constraint Creation Methods

You cannot create constraints using the VB API applications. The following methods are not supported. These methods will be supported in a future release:

- `CCpfcDatumPlaneThroughConstraint.Create()`
- `CCpfcDatumPlaneNormalConstraint.Create()`
- `CCpfcDatumPlaneParallelConstraint.Create()`
- `CCpfcDatumPlaneTangentConstraint.Create()`
- `CCpfcDatumPlaneOffsetConstraint.Create()`
- `CCpfcDatumPlaneOffsetCoordSysConstraint.Create()`
- `CCpfcDatumPlaneAngleConstraint.Create()`
- `CCpfcDatumPlaneSectionConstraint.Create()`
- `CCpfcDatumPlaneDefaultXConstraint.Create()`
- `CCpfcDatumPlaneDefaultYConstraint.Create()`
- `CCpfcDatumPlaneDefaultZConstraint.Create()`
- `CCpfcDatumAxisConstraint.Create()`
- `CCpfcDatumAxisDimensionConstraint.Create()`
- `CCpfcDatumPointPlacementConstraint.Create()`
- `CCpfcDatumPointDimensionConstraint.Create()`
- `CCpfcDatumCsysOriginConstraint.Create()`
- `CCpfcDatumCsysOrientMoveConstraint.Create()`
- `CCpfcDatumCsysDimensionConstraint.Create()`

Support for Feature Subclasses

From Creo Parametric 3.0 onward, only the following subclasses of the class `pfcFeature` are supported:

- `pfcComponentFeat`
- `pfcCoordSysFeat`
- `pfcCurveFeat`
- `pfcDatumAxisFeat`

- `pfcDatumPlaneFeat`
- `pfcDatumPointFeat`
- `pfcRoundFeat`

If your applications check the type of feature using the type of class, then you must update your existing code to use the method `IpfcFeature.FeatType`. The method `IpfcFeature.FeatType` returns the type of feature.

Support for Multi-CAD Assemblies

Multi-CAD assemblies are not supported in VB API. Refer to the respective user's guide for information on multi-CAD assemblies.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0.

Application Information: Compatibility

New Function	Description
<code>IpfcSession.GetAppInfo</code> <code>IpfcSession.SetAppInfo</code>	Gets and sets information related to compatibility in an application.
<code>CCpfcAppInfo.Create</code> <code>IpfcAppInfo.GetCompatibility</code> <code>IpfcAppInfo.SetCompatibility</code>	Gets and sets the compatibility value for the specified application.

Drawings

New Function	Description
<code>IpfcTableOwner.RetrieveTableByOrigin</code>	Retrieves and places a drawing table in the drawing at the specified point. The origin of the table is positioned at the specified point.

Simplified Representation

New Function	Description
<code>CCpfcSimpRepDefaultEnvelope ()</code>	Creates an object that represents the component in default envelope type of simplified representation.
<code>CCpfcSimpRepBoundingBox.Create ()</code>	Creates an object that represents the component in boundary box type of simplified representation.
<code>CCpfcSimpRepLightWeightGraphics.Create ()</code>	Creates an object that represents the component in lightweight type of simplified representation.

Superseded Functions

This section describes the superseded functions for VB API for Creo Parametric 4.0.

Support for Windchill ProductPoint

From Creo Parametric 3.0 F000 onward, the module `WPP` along with its methods has been obsoleted as Windchill ProductPoint is no longer supported.

Miscellaneous Technical Changes

The following changes in Creo Parametric 4.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Configuration Flag to Include Annotations During Export of Creo Parametric Models

From Creo Parametric 3.0 onward, a new configuration flag `EpfCEXPORT_INCLUDE_ANNOTATIONS` has been added. The flag includes annotations during the export of Creo Parametric model to the specified format.

Display Style for Views

From Creo Parametric 3.0 M010 onward, a new display type `EpfCDISPSTYLE_SHADED_WITH_EDGES` has been added to the enumerated data type `EpfCDisplayStyle`. This option allows you to display the model as a shaded solid along with its edges.

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Summary of Technical Changes for Creo 4.0 M030

Critical Technical Changes	14
New Functions	14

This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M030 and VB API. It also lists the new and superseded functions for this release.

Critical Technical Changes

This section describes the changes in Creo Parametric 4.0 M030 and VB API that might require alteration of existing VB API applications.

Exporting to Other File Formats Using the Export Profile Option

It is recommended to use the method `IpfcModel::ExportIntf3D` to export Creo Parametric models to other file formats. The export options must be set using the export profile option in Creo Parametric.

The following interfaces along with their methods will be deprecated in a future release of Creo Parametric:

- `IpfcExport3DInstructions`
- `IpfcACIS3DExportInstructions`
- `IpfcCATIAModel3DExportInstructions`
- `IpfcCATIASession3DExportInstructions`
- `IpfcCatiaPart3DExportInstructions`
- `IpfcCatiaProduct3DExportInstructions`
- `IpfcCatiaCGR3DExportInstructions`
- `IpfcDXF3DExportInstructions`
- `IpfcDWG3DExportInstructions`
- `IpfcIGES3DNewExportInstructions`
- `IpfcJT3DExportInstructions`
- `IpfcParaSolid3DExportInstructions`
- `IpfcSTEP3DExportInstructions`
- `IpfcSWPart3DExportInstructions`
- `IpfcSWAsm3DExportInstructions`
- `IpfcUG3DExportInstructions`
- `IpfcVDA3DExportInstructions`

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M030.

Data Exchange

New Function	Description
IpfcModel::ExportIntf3D	Exports a Creo Parametric model to the specified output format using the default export profile.

Features

New Function	Description
IpfcBaseSession.QueryFeatureEdit	Lists all the features that are currently being edited by the Edit Definition command.

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Summary of Technical Changes for Creo 4.0 M040

New Functions	17
Miscellaneous Technical Changes	17

This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M040 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M040.

Simplified Representation

New Function	Description
<code>CCpfcSimpRepDefaultEnvelope()</code>	Creates an object that represents the component in default envelope type of simplified representation.
<code>CCpfcSimpRepBoundingBox.Create()</code>	Creates an object that represents the component in boundary box type of simplified representation.
<code>CCpfcSimpRepLightWeightGraphics.Create()</code>	Creates an object that represents the component in lightweight type of simplified representation.

Miscellaneous Technical Changes

The following changes in Creo Parametric 4.0 M040 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Plotting a Layout Without BorderSize

From Creo Parametric 4.0 M040 onward, you can plot a layout without adding a border on the paper. To plot, use the following paper sizes defined in the enumerated data type `EpfcPlotPaperSize`:

- `EpfcCEEMPTYPLOT`—The paper size is 22.5 x 36 in
- `EpfcCEEMPTYPLOT_MM`—The paper size is 625 x 1000 mm

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Summary of Technical Changes for Creo 4.0 M050

New Functions	19
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M050 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M050.

View Owner

New Function	Description
IpfcViewOwner. GetCurrentViewTransform	Retrieves the transformation for a model in the current view.
IpfcViewOwner. SetCurrentViewTransform	Sets the transformation for a model in the current view.
IpfcViewOwner.CurrentViewRotate	Rotates the object in the current view relative to X, Y, or Z axis.

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Summary of Technical Changes for Creo 4.0 M060

New Functions	21
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M060 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M060.

Selection

New Function	Description
CMpfcSelect.CreateModelSelection	Creates an <code>IpfcSelection</code> object, based on an <code>IpfcModel</code> object.

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Summary of Technical Changes for Creo 4.0 M070

New Functions	23
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M070 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M070.

Drawings

New Function	Description
Details Notes Operations	
IpfcDetailNoteItem. KeepArrowTypeAsIs()	Allows you to keep arrow type of the leader note as it is, after a note is modified.

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Summary of Technical Changes for Creo 4.0 M100

Miscellaneous Technical Changes	25
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M100 and VB API. It also lists the new and superseded functions for this release.

Miscellaneous Technical Changes

The following changes in Creo Parametric 4.0 M100 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Automatic Representation

From Creo Parametric 4.0 M100 onward, the following values are added to the enumerated data type `EpfcSimpRepActionType`:

- `EpfcSIMPREP_AUTO`—Use the automatic representation of the component.

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Summary of Technical Changes for Creo 4.0 M110

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M110 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M110.

Drawings

New Function	Description
IpfcGeneralViewCreateInstructions. ViewScale	Assigns a scale to the created general drawing view. This value is optional, if not assigned, the default drawing scale is used.

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Summary of Technical Changes for Creo 4.0 M140

New Functions	29
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M140 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M140.

Assembly

New Function	Description
IpfcComponentFeat. GetConstraintsWithCompPath()	Retrieves the constraints for a given assembly component and the component path.

Interface

New Function	Description
IpfcBaseSession.ExportProfileLoad()	Loads the specified profile for export.

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Summary of Technical Changes for Creo 4.0 M150

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 4.0 M150 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 4.0 M150.

Graphics

New Function	Description
CMpfcBase.MakeMatrixOrthonormal()	Converts a non-orthonormal matrix to an orthonormal matrix with the specified scaling factor.

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Technical Summary of Changes for Creo 5.0.0.0

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Superseded Functions	33

This chapter describes the critical and miscellaneous technical changes in Creo Parametric 5.0.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 5.0.0.0.

Data Exchange

New Function	Description
IpfcModel.ExportIntf3D	Exports a Creo Parametric model to the specified output format using the default export profile.

View Owner

New Function	Description
IpfcViewOwner. GetCurrentViewTransform	Retrieves the transformation for a model in the current view.
IpfcViewOwner. SetCurrentViewTransform	Sets the transformation for a model in the current view.
IpfcViewOwner.CurrentViewRotate	Rotates the object in the current view relative to X, Y, or Z axis.

Superseded Functions

This section describes the superseded functions for VB API for Creo Parametric 5.0.0.0.

Data Exchange

Superseded Function	New Function
IpfcExport3DInstructions. Configuration	IpfcModel.ExportIntf3D
IpfcExport3DInstructions. ReferenceSystem	
IpfcExport3DInstructions.Geometry	
IpfcExport3DInstructions. IncludedEntities	
IpfcExport3DInstructions.LayerOptions	
CCpfcSTEP3DExportInstructions. Create	
CCpfcVDA3DExportInstructions. Create	
CCpfcIGES3DNewExportInstructions. Create	
CCpfcCATIAModel3DExportInstruc- tions.Create	
CCpfcACIS3DExportInstructions. Create	
CCpfcCatiaPart3DExportInstructions. Create	
CCpfcCatiaProduct3DExportInstruc- tions.Create	
CCpfcCatiaCGR3DExportInstructions. Create	
CCpfcDXF3DExportInstructions. Create	
CCpfcDWG3DExportInstructions. Create	
CCpfcJT3DExportInstructions.Create	
CCpfcParaSolid3DExportInstructions. Create	
CCpfcUG3DExportInstructions.Create	

From Creo Parametric 5.0.0.0 F000, the following interfaces along with their methods have been deprecated. Use the method `IpfcModel.ExportIntf3D` instead to export Creo Parametric models to other file formats. The export options must be set using the export profile option in Creo Parametric.

- `IpfcExport3DInstructions`
- `IpfcACIS3DExportInstructions`
- `IpfcCATIAModel3DExportInstructions`
- `IpfcCATIASession3DExportInstructions`
- `IpfcCatiaPart3DExportInstructions`
- `IpfcCatiaProduct3DExportInstructions`
- `IpfcCatiaCGR3DExportInstructions`
- `IpfcDXF3DExportInstructions`
- `IpfcDWG3DExportInstructions`
- `IpfcIGES3DNewExportInstructions`
- `IpfcJT3DExportInstructions`
- `IpfcParaSolid3DExportInstructions`
- `IpfcSTEP3DExportInstructions`
- `IpfcSWPart3DExportInstructions`
- `IpfcSWAsm3DExportInstructions`
- `IpfcUG3DExportInstructions`
- `IpfcVDA3DExportInstructions`

Detail Text

Superseded Function	New Function
<code>IpfcDetailText.TextHeight</code>	<code>IpfcDetailText.TextStyle</code> <code>IpfcAnnotationTextStyle.Height</code>
<code>IpfcDetailText.TextWidthFactor</code>	<code>IpfcDetailText.TextStyle</code> <code>IpfcAnnotationTextStyle.Width</code>
<code>IpfcDetailText.TextSlantAngle</code>	<code>IpfcDetailText.TextStyle</code> <code>IpfcAnnotationTextStyle.SlantAngle</code>
<code>IpfcDetailText.TextThickness</code>	<code>IpfcDetailText.TextStyle</code> <code>IpfcAnnotationTextStyle.Thickness</code>

Superseded Function	New Function
IpfcDetailText.FontName	IpfcDetailText.TextStyle IpfcAnnotationTextStyle.Font
IpfcDetailText.IsUnderlined	IpfcDetailText.TextStyle IpfcAnnotationTextStyle. IsTextUnderlined

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Technical Summary of Changes for Creo 5.0.2.0

New Functions	38
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 5.0.2.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 5.0.2.0.

Drawings

New Function	Description
Details Notes Operations	
IpfcDetailNoteItem. KeepArrowTypeAsIs()	Allows you to keep arrow type of the leader note as it is, after a note is modified.

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Technical Summary of Changes for Creo 6.0.0.0

New Functions	40
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 6.0.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 6.0.0.0.

Action Listeners

New Function	Description
IpfcActionSource. AddActionListenerWithType()	Adds action listeners to notify you about specified events.

Technical Summary of Changes for Creo 6.0.3.0

New Functions	42
Miscellaneous Technical Changes	42

This chapter describes the critical and miscellaneous technical changes in Creo Parametric 6.0.3.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 6.0.3.0.

Drawings

New Function	Description
<code>IpfcGeneralViewCreateInstructions.ViewScale</code>	Assigns a scale to the created general drawing view. This value is optional, if not assigned, the default drawing scale is used.

Miscellaneous Technical Changes

The following changes in Creo Parametric 6.0.3.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Automatic Representation

From Creo Parametric 6.0.3.0 and later, the following value is added to the enumerated data type `EpfcSimpRepActionType`:

- `EpfcSIMPREP_AUTO`—Use the automatic representation of the component.

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Technical Summary of Changes for Creo 6.0.5.0

New Functions	44
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 6.0.5.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 6.0.5.0.

Interface

New Function	Description
IpfcBaseSession.ExportProfileLoad()	Loads the specified profile for export.

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Technical Summary of Changes for Creo 6.0.6.0

New Functions	46
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 6.0.6.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 6.0.6.0.

Assembly

New Function	Description
IpfcComponentFeat. GetConstraintsWithCompPath()	Retrieves the constraints for a given assembly component and the component path.

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Technical Summary of Changes for Creo 7.0.0.0

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 7.0.0.0.

Drawings

New Function	Description
<code>IpfcGeneralViewCreateInstructions.ViewScale</code>	Assigns a scale to the created general drawing view. This value is optional, if not assigned, the default drawing scale is used.

Materials

New Function	Description
<code>IpfcMaterial.FluidMaterialType</code>	Sets the material type for the fluid properties of the material.

Solid Bodies

New Function	Description
<code>IpfcSolidBody.GetBodyState()</code>	Specifies the state of the body using the enumerated data type <code>EpfcSolidBodyState</code> .
<code>IpfcSolidBody.IsConstruction()</code>	Checks if the body is a construction body.
<code>IpfcSolidBody.ListSurfaces()</code>	Lists all the surfaces in the specified body.
<code>IpfcSolid.GetDefaultBody()</code>	Returns the default body in the specified solid.
<code>IpfcSolidBody.GetFeatures()</code>	Lists the features that are associated with the specified body.
<code>IpfcSolid.GetSurfaceSolidBody()</code>	Returns the body to which the surface belongs.
<code>IpfcSolid.GetEdgeSolidBody()</code>	Returns the body to which the edge belongs.
<code>IpfcSolidBody.GetOutline()</code>	Retrieves the regeneration outline of a solid body, with respect to the base coordinate system orientation.
<code>IpfcSolidBody.IsSheetmetal()</code>	Checks if the specified body is an

New Function	Description
	active sheetmetal body.
<code>IpfcSolidBody.GetDensity()</code>	Determines the density of the body.
<code>IpfcSolidBody.GetMassProperty()</code>	Returns the mass properties of a body in the specified coordinate system.

Miscellaneous Technical Changes

The following changes in Creo Parametric 7.0.0.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Automatic Representation

The value `EpfcSIMPREP_AUTO` is added to the enumerated data type `EpfcSimpRepActionType` which indicates the use of automatic representation of the component

Support for Fluid as Material Type

The value `EpfcMTL_FLUID` is added to the enumerated data type `EpfcMaterialType` which indicates geometric bodies can now have material type as fluid.

Change in behavior of the property `IpfcFeature.FeatType`

Earlier the property `IpfcFeature.FeatType` returned the error `IpfcXInvalidEnumValue` for Belt feature. In Creo Parametric 7.0.0.0 and later, this behavior is fixed. To support the Belt feature in the VB API application, the following enumerated types are updated with the new values:

- `EpfcComponentConstraintType`
 - `EpfcASM_CONSTRAINT_EXPLICIT`
- `EpfcFeatureType`
 - `EpfcFEAT_HULL_BLOCK`
 - `EpfcFEAT_HULL_BLOCK_DEF`
 - `EpfcFEAT_PM_BELT`

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Technical Summary of Changes for Creo 7.0.1.0

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.1.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 7.0.1.0.

Assembly

New Function	Description
IpfcComponentFeat. GetConstraintsWithCompPath()	Retrieves the constraints for a given assembly component and the component path.

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Technical Summary of Changes for Creo 7.0.2.0

New Functions	53
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.2.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 7.0.2.0.

Interface

New Function	Description
IpfcBaseSession.ExportProfileLoad()	Loads the specified profile for export.

Solids

New Function	Description
IpfcSolid.GetMassPropertyWithDensity() ()	Calculates the mass properties of a part or an assembly in the specified coordinate system.
IpfcRegenInstructions. ResolveModeRegen	Allows regeneration of a solid in resolve mode.

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Technical Summary of Changes for Creo 7.0.3.0

New Functions	55
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.3.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 7.0.3.0.

Features

New Function	Description
IpfcFeature.GroupDirectHeader	Retrieves the direct header of the group.

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Technical Summary of Changes for Creo 7.0.4.0

New Functions	57
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.4.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 7.0.4.0.

Graphics

New Function	Description
CMpfcBase.MakeMatrixOrthonormal()	Converts a non-orthonormal matrix to an orthonormal matrix with the specified scaling factor.

Solid

New Function	Description
IpfcAssembly. GetMassPropertyByCompPath()	Calculates the mass properties of a solid that is referenced by the specified coordinate system selection, using the respective component paths.

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Technical Summary of Changes for Creo 7.0.6.0

Miscellaneous Technical Changes	59
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 7.0.6.0 and VB API. It also lists the new and superseded functions for this release.

Miscellaneous Technical Changes

The following changes in Creo 7.0.6.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Spline Surfaces with 2 Derivatives

The value `EpfcSURFACE_SPL2DER` is added to the enumerated data type `EpfcSurfaceType`, which provides support for spline surfaces with 2 derivatives.

Support for Datum Curve End in Attachment Selection

The value `EpfcITEM_CRV_START` and `EpfcITEM_CRV_END` is added to the enumerated data type `EpfcModelItemType`, which supports selection of specific dimension attachment.

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Technical Summary of Changes for Creo 8.0.0.0

New Functions	61
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 8.0.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 8.0.0.0.

Features

New Function	Description
IpfcFeature.GroupDirectHeader	Retrieves the direct header of the group.

Graphics

New Function	Description
CMpfcBase.MakeMatrixOrthonormal()	Converts a non-orthonormal matrix to an orthonormal matrix with the specified scaling factor.

Interface

New Function	Description
IpfcBaseSession.ExportProfileLoad()	Loads the specified profile for export.

Solids

New Function	Description
IpfcSolid.GetMassPropertyWithDensity() ()	Calculates the mass properties of a part or an assembly in the specified coordinate system.
IpfcRegenInstructions. ResolveModeRegen	Allows regeneration of a solid in resolve mode.

Technical Summary of Changes for Creo 8.0.1.0

New Functions	63
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 8.0.1.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 8.0.1.0.

Solids

New Function	Description
IpfcAssembly. GetMassPropertyByCompPath()	Calculates the mass properties of a solid that is referenced by the specified coordinate system selection, using the respective component paths.

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Technical Summary of Changes for Creo 8.0.2.0

Miscellaneous Technical Changes	65
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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 8.0.2.0 and VB API. It also lists the new and superseded functions for this release.

Miscellaneous Technical Changes

The following changes in Creo 8.0.2.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Spline Surfaces with 2 Derivatives

The value `EpfCSURFACE_SPL2DER` is added to the enumerated data type `EpfcSurfaceType`, which provides support for spline surfaces with 2 derivatives.

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Technical Summary of Changes for Creo 8.0.3.0

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Miscellaneous Technical Changes	67

This chapter describes the critical and miscellaneous technical changes in Creo Parametric 8.0.3.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 8.0.3.0.

Action Listeners

New Function	Description
<code>IpfcSelectionBufferListener.OnAfterSelBufferChange()</code>	This method is called after the selection buffer is changed. This method is available by calling the method <code>IpfcActionSource.AddActionListenerWithType</code> with the value of the notify type as <code>EpfcSELBUFFER_CHANGE_POST</code> .

Miscellaneous Technical Changes

The following changes in Creo 8.0.3.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Datum Curve End in Attachment Selection

The value `EpfcITEM_CRV_START` and `EpfcITEM_CRV_END` is added to the enumerated data type `EpfcModelItemType`, which supports selection of specific dimension attachment.

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Technical Summary of Changes for Creo 9.0.0.0

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 9.0.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 9.0.0.0.

Action Listeners

New Function	Description
<code>IpfcSelectionBufferListener.OnAfterSelBufferChange()</code>	This method is called after the selection buffer is changed. This method is available by calling the method <code>IpfcActionSource.AddActionListenerWithType</code> with the value of the notify type as <code>EpfcSELBUFFER_CHANGE_POST</code> .

Miscellaneous Technical Changes

The following changes in Creo 9.0.0.0 can affect the functional behavior of VB API. PTC does not anticipate that these changes cause critical issues with existing VB API applications.

Support for Spline Surfaces with 2 Derivatives

The value `EpfcSURFACE_SPL2DER` is added to the enumerated data type `EpfcSurfaceType`, which provides support for spline surfaces with 2 derivatives.

Support for Datum Curve End in Attachment Selection

The value `EpfcITEM_CRV_START` and `EpfcITEM_CRV_END` is added to the enumerated data type `EpfcModelItemType`, which supports selection of specific dimension attachment.

Obsolete Function

In Creo Parametric 9.0.0.0 and later, the property `IpfcModelDescriptor.Host` is obsoleted and will not have any support.

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 10.0.6.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 10.0.6.0.

Features

New Function	Description
IpfcFeature.GetPatternByType()	Returns the pattern of the feature.
IpfcFeature.PatternStatus	Returns the pattern status of the feature.
IpfcFeature.GroupStatus	Returns the group status of the feature.
IpfcFeature.GroupPatternStatus	Returns the group pattern status of the feature.

Full Version of Creo® Parametric VB API Release Notes

To see a full version of the *Creo® Parametric VB API Release Notes*, visit the page [Creo® Parametric VB API Release Notes](#). The full version contains information from all the past release notes for VB API.

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Technical Summary of Changes for Creo 11.0.2.0

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 11.0.2.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 11.0.2.0.

Features

New Function	Description
IpfcFeature.GetPatternByType()	Returns the pattern of the feature.
IpfcFeature.PatternStatus	Returns the pattern status of the feature.
IpfcFeature.GroupStatus	Returns the group status of the feature.
IpfcFeature.GroupPatternStatus	Returns the group pattern status of the feature.

Full Version of Creo® Parametric VB API Release Notes

To see a full version of the *Creo® Parametric VB API Release Notes*, visit the page [Creo® Parametric VB API Release Notes](#). The full version contains information from all the past release notes for VB API.

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This chapter describes the critical and miscellaneous technical changes in Creo Parametric 12.4.0.0 and VB API. It also lists the new and superseded functions for this release.

New Functions

This section describes new functions for VB API for Creo Parametric 12.4.0.0.

Features

New Function	Description
IpfcFeature.GetPatternByType()	Returns the pattern of the feature.
IpfcFeature.PatternStatus	Returns the pattern status of the feature.
IpfcFeature.GroupStatus	Returns the group status of the feature.
IpfcFeature.GroupPatternStatus	Returns the group pattern status of the feature.

Full Version of Creo® Parametric VB API Release Notes

To see a full version of the *Creo® Parametric VB API Release Notes*, visit the page [Creo® Parametric VB API Release Notes](#). The full version contains information from all the past release notes for VB API.